

A One-Point Resolvent–Hausdorff Criterion for the Riemann Ξ Function

Lluís Eriksson

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Status and scope. This manuscript does not claim a proof of the Riemann hypothesis. It proves an abstract reduction, develops finite certificate hierarchies, and supplies a finite Lean 4 proof layer with mandatory CI verification. The concrete spectral convergence that would imply RH remains open. Novelty relative to existing Hausdorff-moment criteria also requires specialist review.

Abstract

For $x > 1/4$ define

$$\mathcal{S}_{\Xi}(x) = \frac{1}{2\sqrt{x}} \frac{\xi'}{\xi} \left(\frac{1}{2} + \sqrt{x} \right), \quad \Xi(z) = \xi \left(\frac{1}{2} + iz \right).$$

We isolate a slit-plane extension criterion: if $\mathcal{S}_{\Xi}$ agrees on one real interval with a holomorphic function on $\mathbb{C} \setminus (-\infty, 0]$, then every zero of Ξ is real. Fixing a single $x_0 > 1/4$, we encode the derivative jet by

$$b_n(x_0) = x_0^n \frac{(-1)^n}{n!} \mathcal{S}_{\Xi}^{(n)}(x_0).$$

Using the Hausdorff moment theorem, we show that RH is equivalent to the sequence $(b_n(x_0))_{n \geq 0}$ being a moment sequence of a finite positive measure on $[0, 1]$, or equivalently to all signed forward differences being nonnegative. Under RH the measure is obtained by the resolvent compactification $\gamma^2 \mapsto x_0/(\gamma^2 + x_0)$ of the ordinates of the zeros. Conversely, a Hausdorff representing measure reconstructs a slit-plane extension. We derive finite difference and positive semidefinite certificate hierarchies, an explicit prime-side target with a uniform elementary truncation bound, and a normal-family criterion for positive squared-resolvent approximants. A Lean 4 companion encodes proofs of the finite atomic identities, compactification inequalities, sum-of-squares certificates, and error-budget glue. The decisive open task is a non-circular convergence estimate for a concrete prime-built self-adjoint family.

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1 Introduction

The Riemann hypothesis is the assertion that every non-trivial zero of the zeta function has real part $1/2$ [1, 2, 13]. A classical spectral ambition is to realize the corresponding ordinates as the spectrum of a self-adjoint operator. Recent work has produced prime-built self-adjoint approximants with striking numerical behaviour, while identifying rigorous convergence as the central unresolved step [9, 10].

This paper advocates a positive scalar observable rather than direct convergence of regularized determinants:

$$S_D(x) = \frac{1}{2} \text{Tr}(D^2 + xI)^{-1}.$$

It suppresses high spectral values, is stable under perturbation, and belongs to the Stieltjes class whenever the squared resolvent is trace class. The transformation

$$\lambda \mapsto \frac{x_0}{\lambda + x_0}$$

compactifies a nonnegative spectrum to $[0, 1]$. At a single fixed point x_0 , the entire derivative jet of the target therefore becomes a candidate Hausdorff moment sequence.

Moment formulations of RH already exist; in particular Zhang develops Hausdorff-moment criteria for classes of entire functions and for RH [8]. Accordingly, the present manuscript does not assert that “moments imply RH” is new. Its proposed contribution is the integrated package: a squared-resolvent target, one-point derivative jet, slit-plane reconstruction, prime-side truncation, finite semidefinite certificates, and formal verification. Whether the precise one-point formulation is novel is a literature question, not a theorem.

Contributions. The paper establishes the following chain.

- (i) A slit-plane extension of $\mathcal{S}_\Xi$ excludes non-real zeros of Ξ .

- (ii) For every fixed $x_0 > 1/4$, RH is equivalent to a Hausdorff moment condition on the derivatives of $\mathcal{S}_\Xi$ at x_0 .
- (iii) The moment condition yields scalar finite-difference and finite PSD tests.
- (iv) The arithmetic target has an absolutely convergent prime expansion and an explicit elementary tail bound.
- (v) Positive Stieltjes approximants with one-point control need only converge on one real interval to imply RH.
- (vi) A Lean 4 artifact encodes the finite algebraic substrate without project axioms and makes compilation an explicit CI gate.

2 The completed zeta function and the resolvent target

Define

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s), \quad \Xi(z) = \xi\left(\frac{1}{2} + iz\right). \quad (1)$$

The function ξ is entire and satisfies $\xi(s) = \xi(1-s)$; hence Ξ is entire, even, and real on the real axis. The non-trivial zero $\rho = \beta + i\gamma$ of ζ corresponds to the zero

$$z = \gamma - i\left(\beta - \frac{1}{2}\right)$$

of Ξ . Thus

$$\text{RH} \iff \text{all zeros of } \Xi \text{ are real.} \quad (2)$$

For $x > 1/4$, set

$$\mathcal{S}_\Xi(x) = \frac{1}{2\sqrt{x}} \frac{\xi'}{\xi} \left(\frac{1}{2} + \sqrt{x} \right). \quad (3)$$

The restriction ensures $\sigma = 1/2 + \sqrt{x} > 1$, where the Euler product is absolutely convergent and $\zeta(\sigma) \neq 0$.

Proposition 2.1 (Spectral representation under RH). *Assume RH and enumerate the positive zeros of Ξ , with multiplicity, by $0 < \gamma_1 \leq \gamma_2 \leq \dots$. Then*

$$\mathcal{S}_\Xi(x) = \sum_{j \geq 1} \frac{1}{\gamma_j^2 + x}, \quad x > 0. \quad (4)$$

In particular $\mathcal{S}_\Xi$ is the Stieltjes transform of the positive discrete measure $\sum_j \delta_{\gamma_j^2}$.

Proof. Because Ξ is even of order one and $\sum_j \gamma_j^{-2} < \infty$, its symmetric canonical product has the form

$$\Xi(z) = \Xi(0) \prod_{j \geq 1} \left(1 - \frac{z^2}{\gamma_j^2} \right).$$

Logarithmic differentiation gives

$$-\frac{\Xi'(z)}{\Xi(z)} = 2z \sum_{j \geq 1} \frac{1}{\gamma_j^2 - z^2}.$$

At $z = iy$, the functional equation implies

$$-\frac{\Xi'(iy)}{\Xi(iy)} = i \frac{\xi'}{\xi} \left(\frac{1}{2} + y \right).$$

Dividing by $2iy$ and putting $x = y^2$ yields (4). Standard zero counting gives convergence locally uniformly away from the negative real cut. $\square$

3 A slit-plane extension criterion

Let

$$\Omega = \mathbb{C} \setminus (-\infty, 0].$$

The quadratic map used below is the key geometric device.

Lemma 3.1. *If $\operatorname{Im} z > 0$, then $-z^2 \in \Omega$.*

Proof. Write $z = a + ib$ with $b > 0$. If $a = 0$, then $-z^2 = b^2 > 0$. If $a \neq 0$, then $\operatorname{Im}(-z^2) = -2ab \neq 0$, so $-z^2$ is not on the non-positive real axis. $\square$

Theorem 3.2 (Slit-plane extension criterion). *Let $I \subset (1/4, \infty)$ be a nonempty open interval. Suppose that a holomorphic function $S : \Omega \rightarrow \mathbb{C}$ satisfies*

$$S(x) = \mathcal{S}_\Xi(x), \quad x \in I.$$

Then RH holds.

Proof. For z in the upper half-plane define

$$M(z) = 2zS(-z^2).$$

By Theorem 3.1, M is holomorphic. If $z = iy$ and $y^2 \in I$, then

$$M(iy) = 2iy\mathcal{S}_\Xi(y^2) = i\frac{\xi'}{\xi} \left(\frac{1}{2} + y \right) = -\frac{\Xi'(iy)}{\Xi(iy)}.$$

The meromorphic identity theorem therefore gives

$$M(z) = -\frac{\Xi'(z)}{\Xi(z)}$$

throughout the upper half-plane away from the zeros of Ξ .

Suppose z_0 were an upper-half-plane zero of multiplicity $m > 0$. Locally $\Xi(z) = (z - z_0)^m g(z)$ with $g(z_0) \neq 0$, and hence

$$-\frac{\Xi'(z)}{\Xi(z)} = -\frac{m}{z - z_0} - \frac{g'(z)}{g(z)}.$$

This has a non-removable pole at z_0 , contradicting the holomorphy of M . Thus no zero lies in the upper half-plane. Conjugation symmetry excludes the lower half-plane, so all zeros are real. By (2), RH follows. $\square$

Remark 3.3. The theorem does not reconstruct every zero, require determinant convergence, or identify an inverse spectral problem. Agreement on one interval is enough because analyticity propagates the logarithmic derivative.

4 The one-point Hausdorff criterion

Fix once and for all $x_0 > 1/4$. Define the normalized derivative jet

$$b_n(x_0) = x_0^n \frac{(-1)^n}{n!} \mathcal{S}_\Xi^{(n)}(x_0), \quad n \geq 0. \tag{5}$$

For a sequence $b = (b_n)$ define the signed forward differences recursively by

$$D^0 b_n = b_n, \quad D^{k+1} b_n = D^k b_n - D^k b_{n+1}.$$

Thus $D^k = (-1)^k \Delta^k$ under the usual convention $\Delta b_n = b_{n+1} - b_n$.

We use the classical Hausdorff moment theorem [4, 5, 6]:

$$b_n = \int_0^1 v^n d\mu(v) \text{ for a finite positive measure } \mu \iff D^k b_n \geq 0 \text{ for all } k, n \geq 0. \quad (6)$$

Theorem 4.1 (One-point resolvent–Hausdorff criterion). *For every fixed $x_0 > 1/4$, the following are equivalent.*

- (a) *RH holds.*
- (b) *The sequence $(b_n(x_0))_{n \geq 0}$ is a Hausdorff moment sequence.*
- (c) *For all $k, n \geq 0$,*

$$D^k b_n(x_0) \geq 0.$$

Proof. Assume RH. By Theorem 2.1, differentiation term by term gives

$$\frac{(-1)^n}{n!} \mathcal{S}_{\Xi}^{(n)}(x_0) = \sum_{j \geq 1} \frac{1}{(\gamma_j^2 + x_0)^{n+1}}.$$

Put

$$u_j = \frac{1}{\gamma_j^2 + x_0}, \quad v_j = \frac{x_0}{\gamma_j^2 + x_0} \in (0, 1).$$

Then

$$b_n(x_0) = \sum_{j \geq 1} u_j v_j^n = \int_0^1 v^n d\mu_{x_0}(v), \quad \mu_{x_0} = \sum_{j \geq 1} u_j \delta_{v_j}.$$

The measure is finite because its total mass is $\mathcal{S}_{\Xi}(x_0)$. This proves (a) $\Rightarrow$ (b), and (b) $\Leftrightarrow$ (c) is (6).

Now assume (b), with representing measure μ . For $w \in \Omega$ define

$$F(w) = \int_0^1 \frac{d\mu(v)}{1 + \frac{w-x_0}{x_0}v}. \quad (7)$$

For $v > 0$, the only zero of the denominator occurs at $w = x_0(1 - v^{-1}) \in (-\infty, 0]$; for $v = 0$ the denominator is one. Standard dominated holomorphic-parameter integration shows that F is holomorphic on Ω .

For h sufficiently small,

$$F(x_0 + h) = \int_0^1 \sum_{n \geq 0} \left(-\frac{h}{x_0}v\right)^n d\mu(v) = \sum_{n \geq 0} (-1)^n \frac{h^n}{x_0^n} b_n(x_0).$$

By (5), this is precisely the Taylor series of $\mathcal{S}_{\Xi}(x_0 + h)$. Consequently F agrees with $\mathcal{S}_{\Xi}$ on a nonempty interval around x_0 . Applying Theorem 3.2 proves RH. $\square$

Corollary 4.2 (Explicit finite differences under RH). *Under RH,*

$$D^k b_n(x_0) = \sum_{j \geq 1} \frac{1}{\gamma_j^2 + x_0} \left(\frac{x_0}{\gamma_j^2 + x_0}\right)^n \left(1 - \frac{x_0}{\gamma_j^2 + x_0}\right)^k \geq 0.$$

5 Finite certificate hierarchies

The infinite family in [Theorem 4.1](#) can be truncated in several falsifiable ways. A single certified negative value of $D^k b_n$ would refute RH. Passing a finite number of tests is evidence about the truncation only, never a proof of RH.

5.1 Finite differences

For integers $K, N \geq 0$, define the scalar test region

$$\mathcal{D}_{K,N} = \{D^k b_n : 0 \leq k \leq K, 0 \leq n \leq N\}.$$

Each quantity can in principle be evaluated from derivatives of the prime-side target, with interval arithmetic used to control truncation and differentiation errors.

5.2 Hankel and localizing forms

Given a moment sequence (b_n) , define

$$H_N = (b_{i+j})_{0 \leq i,j < N}, \quad L_N = (b_{i+j} - b_{i+j+1})_{0 \leq i,j < N}.$$

If $b_n = \int_0^1 v^n d\mu(v)$, then for every $c \in \mathbb{R}^N$,

$$\begin{aligned} c^\top H_N c &= \int_0^1 \left| \sum_{i=0}^{N-1} c_i v^i \right|^2 d\mu(v) \geq 0, \\ c^\top L_N c &= \int_0^1 (1-v) \left| \sum_{i=0}^{N-1} c_i v^i \right|^2 d\mu(v) \geq 0. \end{aligned}$$

Thus

$$H_N \succeq 0, \quad L_N \succeq 0.$$

The Lean source encodes proofs of these facts for finite atomic measures as explicit sum-of-squares certificates, avoiding reliance on numerical eigenvalue computations.

5.3 Resolvent compactification

Let $\lambda_i \geq 0$ be a finite squared spectrum and $x_0 > 0$. Define

$$w_i = \frac{1}{\lambda_i + x_0}, \quad p_i = \frac{x_0}{\lambda_i + x_0} \in [0, 1], \quad b_n^{\text{fin}} = \sum_i w_i p_i^n.$$

Then

$$D^k b_n^{\text{fin}} = \sum_i w_i p_i^n (1 - p_i)^k \geq 0. \tag{8}$$

This identity is the central kernel-checked theorem in the current Lean layer.

6 The arithmetic target and an explicit prime tail

For real $\sigma > 1$, logarithmic differentiation of (1) gives

$$\frac{\xi'}{\xi}(\sigma) = \frac{1}{\sigma} + \frac{1}{\sigma - 1} - \frac{1}{2} \log \pi + \frac{1}{2} \psi\left(\frac{\sigma}{2}\right) - \sum_{m=2}^{\infty} \frac{\Lambda(m)}{m^\sigma}, \tag{9}$$

where $\psi = \Gamma'/\Gamma$ and Λ is the von Mangoldt function. With $y = \sqrt{x}$ and $\sigma = 1/2 + y$,

$$\mathcal{S}_\Xi(x) = \frac{1}{2y} \left[\frac{1}{\sigma} + \frac{1}{\sigma-1} - \frac{1}{2} \log \pi + \frac{1}{2} \psi \left(\frac{\sigma}{2} \right) - \sum_{m=2}^{\infty} \frac{\Lambda(m)}{m^\sigma} \right]. \quad (10)$$

For a cutoff $X \geq 3$, let $P_X(x)$ be the same expression with $m \leq X$.

Proposition 6.1 (Elementary prime-tail bound). *Let $\delta > 0$ and $x \geq (1/2 + \delta)^2$. For $X \geq 3$,*

$$|\mathcal{S}_\Xi(x) - P_X(x)| \leq \frac{X^{-\delta}}{1 + 2\delta} \left(\frac{\log X}{\delta} + \frac{1}{\delta^2} \right). \quad (11)$$

Proof. Here $y \geq 1/2 + \delta$ and $\sigma \geq 1 + \delta$. Since $0 \leq \Lambda(m) \leq \log m$,

$$|\mathcal{S}_\Xi(x) - P_X(x)| \leq \frac{1}{2y} \sum_{m>X} \frac{\log m}{m^{1+\delta}}.$$

For $X \geq 3$ the summand is eventually decreasing, and the standard integral majorant gives

$$\sum_{m>X} \frac{\log m}{m^{1+\delta}} \leq \int_X^\infty \frac{\log t}{t^{1+\delta}} dt = X^{-\delta} \left(\frac{\log X}{\delta} + \frac{1}{\delta^2} \right).$$

Finally $(2y)^{-1} \leq (1 + 2\delta)^{-1}$. □

7 Positive spectral approximants

Let ν_j be positive measures on $[0, \infty)$ satisfying

$$\int_0^\infty \frac{d\nu_j(t)}{t + x_*} < \infty$$

for some $x_* > 0$, and define

$$S_j(w) = \int_0^\infty \frac{d\nu_j(t)}{t + w}, \quad w \in \Omega.$$

For a paired discrete spectrum $\{\pm\gamma_{j,k}\}$ this takes the form

$$S_j(x) = \sum_k \frac{1}{\gamma_{j,k}^2 + x} = \frac{1}{2} \text{Tr}(D_j^2 + xI)^{-1}.$$

Lemma 7.1 (One-point normal-family control). *If $\sup_j S_j(x_*) < \infty$, then (S_j) is locally bounded on Ω .*

Proof. For every compact $K \subset \Omega$ there exists $C_K < \infty$ such that

$$\frac{1}{|t + w|} \leq \frac{C_K}{t + x_*}, \quad t \geq 0, w \in K.$$

Integration yields $|S_j(w)| \leq C_K S_j(x_*)$, uniformly in j . □

Theorem 7.2 (Abstract prime-resolvent criterion). *Let $I \subset (1/4, \infty)$ be a nonempty open interval. Suppose the S_j above satisfy*

$$\sup_j S_j(x_*) < \infty \quad \text{and} \quad S_j(x) \longrightarrow \mathcal{S}_\Xi(x) \quad (x \in I).$$

Then RH holds.

Proof. By Theorem 7.1 and Montel's theorem, a subsequence converges uniformly on compact subsets of Ω to a holomorphic function S . Pointwise convergence on I forces $S = \mathcal{S}_\Xi$ there. The conclusion follows from Theorem 3.2. $\square$

Corollary 7.3 (Prime-resolvent error formulation). *Fix $\delta > 0$ and a compact interval $I_{\delta,B} \subset ((1/2 + \delta)^2, \infty)$. Let $X_j \rightarrow \infty$. If*

$$\sup_{x \in I_{\delta,B}} |S_j(x) - P_{X_j}(x)| \rightarrow 0$$

and the one-point bound in Theorem 7.2 holds, then RH holds.

Proof. Combine the assumed comparison with the uniform bound (11), then apply Theorem 7.2. $\square$

8 A three-part error budget

For a concrete operator construction, the useful decomposition is

$$|S_{\text{spec}} - \mathcal{S}_\Xi| \leq |S_{\text{spec}} - S_{\text{model}}| + |S_{\text{model}} - P_X| + |P_X - \mathcal{S}_\Xi|.$$

The three terms represent:

- (1) spectral/Galerkin or state-alignment error;
- (2) model normalization and transform-convention error;
- (3) the explicit arithmetic tail in Theorem 6.1.

Only the third term is closed here in a source-independent manner. For the prime-built operators of [9], the first two require a precise audit of domains, Fourier/Mellin conventions, spectral gaps, and trace-class estimates.

The finite-dimensional resolvent identity

$$(A^2 + xI)^{-1} - (B^2 + xI)^{-1} = (A^2 + xI)^{-1}(B^2 - A^2)(B^2 + xI)^{-1}$$

suggests the trace estimate

$$\left| \text{Tr} \left((A^2 + xI)^{-1} - (B^2 + xI)^{-1} \right) \right| \leq x^{-2} \|A^2 - B^2\|_1,$$

whenever the indicated trace-class hypotheses hold. Establishing a useful version for the concrete unbounded models is a substantive analytic task, not formal bookkeeping.

9 Numerical illustration and exact toy certificates

The accompanying scripts compare (3), the prime truncation (10), and a finite sum over the first critical-line zeros. The default run uses 40 zeros, prime-power cutoff 20,000, and 70 points on $[1, 25]$. These choices are intentionally modest: the goal is to test conventions and the artifact pipeline, not to provide new evidence for RH.

The exact certificate script uses

$$x_0 = 1, \quad \lambda \in \{1, 4, 9\},$$

so that

$$p \in \left\{ \frac{1}{2}, \frac{1}{5}, \frac{1}{10} \right\}, \quad w = p.$$

It exports rational moments, signed differences, Hankel and localizing matrices, and exact principal minors. Because the support has three atoms, the Hankel rank stabilizes at three; zero higher principal minors are therefore expected.

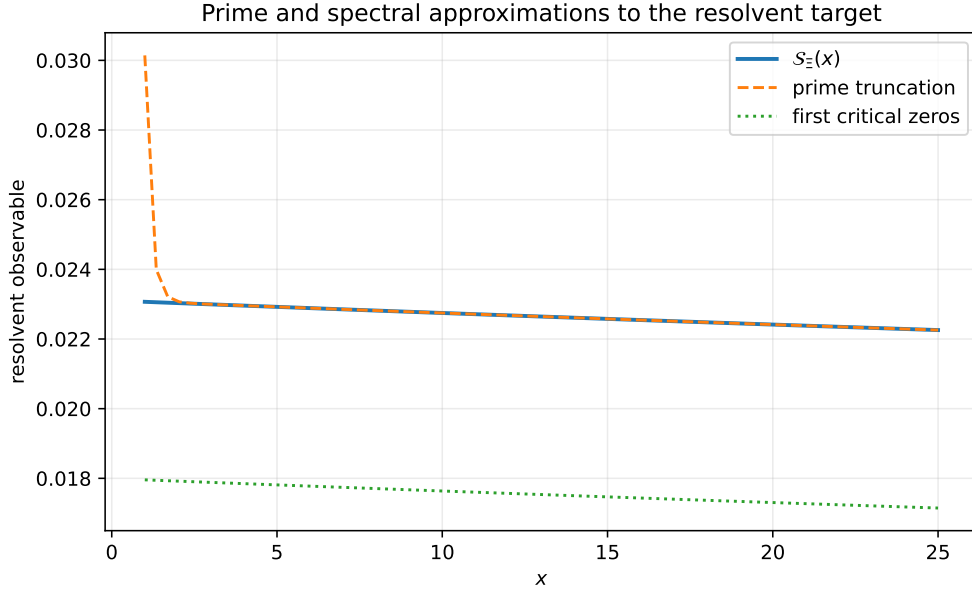


Figure 1: Numerical comparison of the completed-zeta target, a truncated von Mangoldt expansion, and a partial spectral sum. The visible spectral offset is the omitted positive tail, not a contradiction.

10 Lean 4 formalization

The artifact is written for Lean 4 and Mathlib [11, 12], pinned to Lean/Mathlib release v4.31.0. The current source-complete layer is deliberately finite. The construction container used for this archive did not contain Lean, so compilation remains an explicit CI release check rather than a locally claimed success.

Module	Certified content
Basic	Atomic moments, signed forward differences, complete-monotonicity predicate.
HausdorffFinite	Exact identity (8) for finite atoms.
ResolventCompactification	Positivity and $[0, 1]$ support of compactified spectral atoms.
FiniteCertificates	Hankel and localizing sum-of-squares nonnegativity.
ErrorBudget	Deterministic three-term triangle inequality.
Oracle	Kernel axiom audit via <code>#print axioms</code> .

Table 1: Current Lean 4 source proof boundary.

The headline finite theorem has the schematic form:

```
theorem hausdorffDiff_atomicMoment
  (weight point : iota -> Real) (k n : Nat) :
  hausdorffDiff k (atomicMoment weight point) n =
  sum i, weight i * point i ^ n * (1 - point i) ^ k
```

The repository CI is configured to run a full build, the oracle file, a placeholder scan, exact rational certificate tests, and the paper build. No theorem in the current Lean source assumes RH.

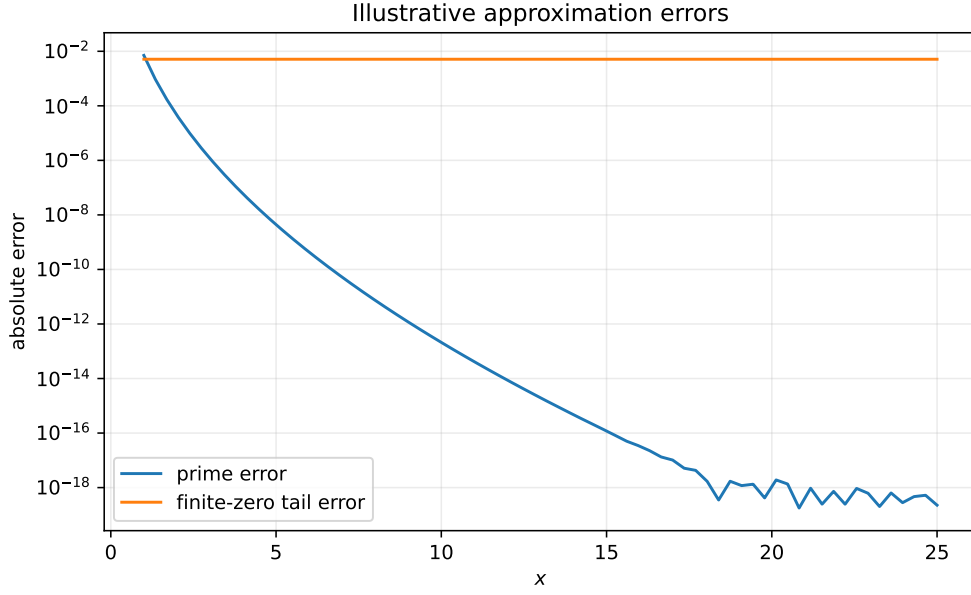


Figure 2: Illustrative absolute errors for the same finite calculation. Neither curve is a certified infinite-dimensional convergence result.

Formalization frontier. The next high-value steps are the exact ξ/Ξ convention bridge, the slit-plane map, local factorization of analytic zeros, the Hausdorff theorem, and the prime-tail estimate. The unbounded operator layer should be introduced only after this abstract chain is closed.

11 Novelty, falsifiability, and publication gates

The following distinctions are essential.

Classical substance. Stieltjes transforms, Hausdorff moments, normal families, and logarithmic-derivative pole arguments are classical. Moment criteria related to RH already appear in the literature, including [8].

Potentially publishable package. A complete Lean formalization of Theorems 3.2, 4.1 and 6.1, combined with finite certified tests and a careful literature comparison, would be a defensible formalization paper even without a new spectral convergence estimate.

Substantive analytic advance. A new unconditional bound for the Galerkin, ground-state alignment, model, or trace-resolvent error of a concrete prime-built family would materially strengthen the project.

RH claim. Only a full, non-circular convergence theorem satisfying all domain, multiplicity, trace, and normalization obligations would support a claim of proving RH. Finite numerical agreement, no matter how accurate, is insufficient.

12 Conclusion

The squared-resolvent viewpoint turns a global zero-location problem into a sequence of separable tasks. The key structural equivalence is

$$\text{RH} \iff \left\{ x_0^n \frac{(-1)^n}{n!} \mathcal{S}_{\Xi}^{(n)}(x_0) \right\}_{n \geq 0} \text{ is a Hausdorff moment sequence}$$

for any fixed $x_0 > 1/4$. The forward direction compactifies the squared ordinates of the zeros; the reverse direction uses a representing measure to build a holomorphic function on the slit plane, where the logarithmic derivative cannot hide a non-real zero.

The resulting finite differences and semidefinite forms provide exact, falsifiable tests. The prime expansion supplies a directly computable arithmetic target. Positive spectral approximants need only be controlled at one point and converge on one real interval. What remains difficult is not obscured: one must prove a uniform, non-circular comparison for a concrete prime-built self-adjoint family. The repository is designed so that this open frontier is visible in the type and dependency structure rather than disguised as an assumption named “convergence.”

A Finite atomic identity

For completeness, let

$$b_n = \sum_{i=1}^m w_i p_i^n.$$

Induction on k proves

$$D^k b_n = \sum_{i=1}^m w_i p_i^n (1 - p_i)^k.$$

Indeed, the step from k to $k + 1$ is

$$\begin{aligned} D^{k+1} b_n &= D^k b_n - D^k b_{n+1} \\ &= \sum_i w_i p_i^n (1 - p_i)^k - \sum_i w_i p_i^{n+1} (1 - p_i)^k \\ &= \sum_i w_i p_i^n (1 - p_i)^{k+1}. \end{aligned}$$

This proof is mirrored directly in `PrimeResolvent/HausdorffFinite.lean`.

B Reproducibility commands

From a fresh clone:

```
lake exe cache get
lake build
lake env lean PrimeResolvent/Oracle.lean
python3 -m pip install -r requirements.txt
python3 -m pytest -q
python3 scripts/exact_atomic_certificate.py --output-dir data/certificates
python3 scripts/numerical_demo.py --output-dir data/demo
./scripts/build_paper.sh
```

The release manifest records SHA-256 hashes of every distributed source and generated artifact. The numerical scripts are explicitly marked non-probative; exact toy certificates use rational arithmetic.

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